

CE-FS Series

105°C Standard

Feature

Solvent proof (within 2 minutes).

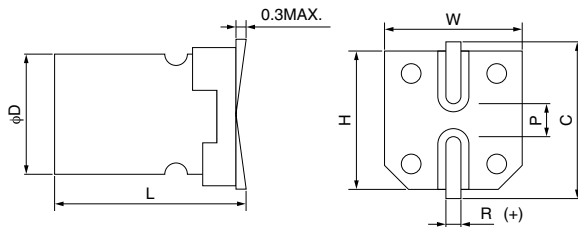


Specifications

Items		Specifications							
Rated voltage (V)		6.3	10	16	25	35	50	63	100
Category temperature range (°C)		-55 to +105							-40 to +105
Capacitance tolerance (%)		±20							(120Hz/20°C)
Tangent of loss angle (tanδ) (MAX.) (120Hz/20°C)	φ4 to φ6.3	0.24	0.20	0.16	0.14	0.12	0.10	0.12	0.10
	φ8 to φ16	0.28	0.24	0.20	0.16	0.14	0.12	0.12	0.10
		When rated capacitance exceeds 1000µF, add 0.02 to the value above for each 1000µF increase.							
Leakage current (L.C.) (µA/after 2min.) (MAX.)		The greater value of either 0.01CV or 3							
Impedance (120Hz) ratio at low temperature (MAX.)	Z-40°C/Z20°C	3	3	2	2	2	2	2	3
	Z-55°C/Z20°C	8	5	4	3	3	3	3	-
Endurance 105°C rated voltage applied	Test	2000hrs. (φD ≤ 6.3, φ10 x 7.7 : 1000hrs.)							
	ΔC/C	Within ± 25% of the initial value							
	tanδ	≤ Twice the initial specified value							
	L.C.	≤ The initial specified value							

Dimensions

(Unit : mm)



A pressure relief vent is attached to products over φD=8

D+0.5MAX.	L	W ±0.2	H ±0.2	C ±0.2	R	P ±0.2
4	5.4 ^{+0.1} _{-0.2}	4.3	4.3	5.0	0.5 to 0.8	1.0
5	5.4 ^{+0.1} _{-0.2}	5.3	5.3	6.0	0.5 to 0.8	1.4
6.3	5.4 ^{+0.1} _{-0.2}	6.6	6.6	7.3	0.5 to 0.8	2.2
4	6.0 ±0.3	4.3	4.3	5.0	0.5 to 0.8	1.0
6.3	6.0 ±0.3	6.6	6.6	7.3	0.5 to 0.8	2.2
6.3	7.7 ±0.3	6.6	6.6	7.3	0.5 to 0.8	2.2
8	10.2 ±0.3	8.3	8.3	9.0	0.7 to 1.0	3.2
10	7.7 ±0.3	10.3	10.3	11.0	1.1 to 1.4	4.6
10	10.2 ±0.3	10.3	10.3	11.0	1.1 to 1.4	4.6
12.5	13.5 ±0.5	12.8	12.8	13.5	1.1 to 1.4	4.6
16	16.5 ±0.5	16.3	16.3	17.3	1.8 to 2.1	7.0

Size List, Rated Ripple Current

µF \ V	6.3	10	16	25	35	50	63	100
0.1 to 0.47						4 x 5.4 10.7 to 3.5	4 x 5.4 10.7 to 3.5	
1.0						4 x 5.4 7	4 x 5.4 7	4 x 6.0 7
2.2						4 x 5.4 11	4 x 5.4 11	6.3 x 6.0 14
3.3						4 x 5.4 13	5 x 5.4 14	6.3 x 6.0 20
4.7				4 x 5.4 13	4 x 5.4 14	5 x 5.4 16	5 x 5.4 16	6.3 x 6.0 25
10			4 x 5.4 18	5 x 5.4 20	5 x 5.4 21	6.3 x 5.4 24	6.3 x 5.4 24	6.3 x 7.7 35
22	4 x 5.4 22	5 x 5.4 25	5 x 5.4 27	6.3 x 5.4 36	6.3 x 5.4 38	6.3 x 6.0 42	6.3 x 7.7 49	8 x 10.2 84
33	5 x 5.4 27	5 x 5.4 30	6.3 x 5.4 40	6.3 x 5.4 44	6.3 x 6.0 42	6.3 x 7.7 60	8 x 10.2 112	10 x 10.2 133
47	5 x 5.4 33	6.3 x 5.4 41	6.3 x 5.4 48	6.3 x 6.0 48	6.3 x 6.0 49	6.3 x 7.7 63	8 x 10.2 119	12.5 x 13.5 240
68							8 x 10.2 126	12.5 x 13.5 245
82						10 x 7.7 140		
100	6.3 x 5.4 50	6.3 x 5.4 53	6.3 x 5.4 60	6.3 x 7.7 91	6.3 x 7.7 84	8 x 10.2 140	10 x 10.2 196	16 x 16.5 490
150		6.3 x 6.0 62	6.3 x 7.7 95	8 x 10.2 140	8 x 10.2 155			16 x 16.5 500
				10 x 7.7 155	10 x 7.7 155			
220	6.3 x 6.0 67	6.3 x 7.7 105	6.3 x 7.7 105	8 x 10.2 175	8 x 10.2 190	10 x 10.2 220	12.5 x 13.5 287	
				10 x 7.7 175				
330	6.3 x 7.7 105	8 x 10.2 195	8 x 10.2 195	8 x 10.2 220	10 x 10.2 300	12.5 x 13.5 365		
			10 x 7.7 195					
390						12.5 x 13.5 380		
470	8 x 10.2 210	8 x 10.2 210	8 x 10.2 230	10 x 10.2 300	12.5 x 13.5 410		16 x 16.5 630	
		10 x 7.7 210						
680	8 x 10.2 210		10 x 10.2 310		12.5 x 13.5 430			
	10 x 7.7 210							
1000	8 x 10.2 230	10 x 10.2 310		12.5 x 13.5 460		16 x 16.5 655		
1500	10 x 10.2 310		12.5 x 13.5 500		16 x 16.5 740			
2200		12.5 x 13.5 510		16 x 16.5 805				
3300	12.5 x 13.5 520		16 x 16.5 840					
4700		16 x 16.5 880						
6800	16 x 16.5 930							

Model No.

16 CE 470 FS

Series code
Rated capacitance symbol
Type code
Rated voltage

10 CE 470 FSA

Series code
Rated capacitance symbol
Type code
Rated voltage

10 CE 4700 FST

Series code
Rated capacitance symbol
Type code
Rated voltage

Case size ; φD x L (mm)
10 x 7.7 ; CE-FSA series
16 x 16.5 ; CE-FST series

Rated ripple current
mAr.m.s. (120Hz, 105°C)

CE-FS Series

105°C Standard

Mid. and High Voltage

Feature

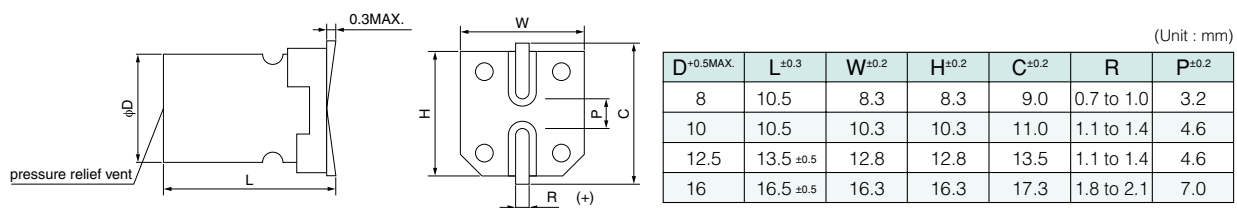
Middle and high voltage product line up of CE-FS series.
It guarantees 2000hrs. over wide temperature range
(-40 to +105°C).



Specifications

Items		Specifications			
Rated voltage (V)		160	200	250	400
Category temperature range (°C)		-40 to +105			
Capacitance tolerance (%)		±20 (120Hz/20°C)			
Tangent of loss angle (tanδ) (MAX.) (120Hz/20°C)		0.20			0.25
Leakage current (L.C.) (μA/after 2min.) (MAX.)		CV≤1000 ; 0.03CV+15 CV>1000 ; 0.02CV+25			
Impedance (120Hz) ratio at low temperature (MAX.)	Z-25°C/Z20°C	3	3	3	6
	Z-40°C/Z20°C	6	6	6	10
Endurance 105°C rated voltage applied	Test	2000hrs. (φD = 8 : 1000hrs.)			
	ΔC/C	Within ±25% of the initial value			
	tanδ	≤ Twice the initial specified value			
	L.C.	≤ The initial specified value			

Dimensions



Size List, Rated Ripple Current

μF \ V	160		200		250		400	
2.2							8 x 10.5	25
3.3					8 x 10.5	31	10 x 10.5	36
4.7					8 x 10.5	37	10 x 10.5	38
6.8					8 x 10.5	44	12.5 x 13.5	47
10	8 x 10.5	57	10 x 10.5	64	10 x 10.5	64	12.5 x 13.5	57
22	12.5 x 13.5	112	12.5 x 13.5	112	12.5 x 13.5	112	16 x 16.5	115
33	12.5 x 13.5	137	12.5 x 13.5	137	16 x 16.5	150		
47	16 x 16.5	180	16 x 16.5	180	16 x 16.5	180		
68	16 x 16.5	215	16 x 16.5	215				
82	16 x 16.5	235						

Case size ; φD x L (mm)
16 x 16.5 ; CE-FST series

Rated ripple current
mAr.m.s. (120Hz, 105°C)

Model No.

200 CE 10 FS
 — Series code
 — Rated capacitance symbol
 — Type code
 — Rated voltage

200 CE 47 FST
 — Series code
 — Rated capacitance symbol
 — Type code
 — Rated voltage